



Science Institute News

Fall 2026

SI Program Updates

The Learning Actively Mentoring Program (LAMP) continued to expand its impact across the University of Wyoming and beyond during 2025-2026, supporting 33 faculty and instructor Fellows, 90 Learning Assistant appointments, and eight educator learning communities dedicated to advancing innovative active learning experiences. Assessments of the LAMP Fellows Program showed growth across every dimension of participants' teaching philosophies, while poster-based evaluations confirmed that Fellows were successfully translating those ideas into course learning outcomes, pedagogies, and assessments. LAs mastered outcomes at the highest levels yet recorded and LA applicant data show that the program is becoming increasingly self-propagating: the share of LAs who say they were inspired by their LA has more than doubled since the early years of the program (2018-2020). The LAMP Director and SI Assessment Specialist collaborated with an external evaluation team from UNC Greensboro to generate a 10-year evaluation of LAMP. This evaluation shows that LAMP participants consistently appreciate that LAMP models the practices that it teaches and responds to participant needs in real time. Ninety-seven percent of LAMP respondents (N=202) agree or strongly agree that "LAMP has meaningfully shaped their teaching, their students' learning, and their professional community". Additional highlights included the launch of a Community College Summer Institute, UW's first Experiential Education Showcase, and expanded collaborations with community colleges across Wyoming and Colorado.

The UW Science Initiative Roadshow celebrated its largest year of outreach yet in Fall 2025–Summer 26, delivering 114

events across 19 Wyoming counties and reaching more than 12,500 students and community members through STEM Days, classroom visits, field experiences, community science events, and on-campus programming. The Roadshow also reached a major statewide milestone, having now provided programming in all 23 Wyoming counties since the program began in 2017. The Roadshow continues to connect UW with schools and communities through hands-on experiences in areas of STEM. Nearly 40 UW undergraduate and graduate Outreach Assistants

IMPORTANT DATES

- [Applications](#) are now open for the 2027-2028 LAMP Fellows Program, "Action Learning with the Machine," which will explore the integration of active learning with artificial intelligence, computation, and machine learning. The soft application deadline is December 16, with all applications due by January 7.
- [Applications](#) for Spring 2027 Learning Assistants (LAs) are also open and are due November 6.
- A WRSP-CURE Expo poster session, exhibiting student research across the programs, will be held Tuesday, December 8th from 5-7 PM in the Science Initiative Building, Room 2010.
- WY Undergraduate Research Coalition (WURC) student club monthly meetings - next meeting is Wed, Oct 14 from 7-8 PM in SIB Room 4004. Please contact Dr. Jamie Crait for membership information.

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SI Program Updates Cont.

helped plan and deliver programming while developing valuable skills in teaching, communication, leadership, and community engagement. Strong partnerships with UW programs, community colleges, state agencies, local organizations, and educators continue to expand the Roadshow's reach and provide meaningful, place-based STEM experiences for learners across Wyoming. The Roadshow has also started a new program, WYOSTEM 360, which welcomes 7-12th grade students from across Wyoming for a STEM visit to the UW campus that aligns with their classroom learning topics, allowing secondary education STEM teachers to link classroom learning with real, Wyoming-relevant science. During the month of November, the program will welcome students from Rock Springs HS, Pine Bluffs HS, and Burlington HS.

The Wyoming Research Scholars Program (WRSP) is starting the 2026-27 academic year with its largest cohort to date, supporting 90 undergraduate scholars engaged in research across STEM disciplines at UW. Continued support from the state, along with partnerships with the Wyoming NASA Space Grant Consortium, Wyoming INBRE, and new funding from the College of Engineering and Physical Sciences, has allowed WRSP to substantially expand opportunities for undergraduate researchers. Approximately 60 new scholars are participating in the WRSP seminar this fall, joining a vibrant, interdisciplinary community of students conducting research alongside UW faculty. Since welcoming just four students in its inaugural 2015 cohort, WRSP has grown into UW's largest undergraduate research program. Scholars are gaining hands-on experience in research labs and groups across campus, with student researchers presenting their work at professional conferences and contributing to peer-reviewed publications. Beyond campus, WRSP students serve as ambassadors for UW research through science and STEM outreach activities across Wyoming, including with the Science Initiative Roadshow.

The Course-Based Undergraduate Research Experiences (CUREs) program has been hard at work recruiting and supporting faculty to develop their classes into CUREs. We

are running the first iteration of General Biology (LIFE 1010 - 400 students this fall) and Plant and Fungal Biology (LIFE 2023 - 50 students this fall) as CUREs along with other smaller courses (including Intro to Ecological Research and Mycology). The goal of CUREs is to expand access to the high impact practice of undergraduate research to more students, and we are just getting started.

Science Institute Research Centers Update and New Research Center, WyoPATH

After a formal review in April 2026, funding for four of the SI's original centers was renewed, including the Center for Energy Materials (C-EM), the Center for Controlled Environment Agriculture (C-CEA), the Center for Quantum Information Science & Engineering (C-QISE), and the Center for Wildlife, Technology & Computing (WyldTech). One new center, WyoPATH, was funded.

WyoPATH was recommended for funding at the end of academic year 2025-2026. The interdisciplinary leadership team, led by Dr. Sarah Kane of Veterinary Sciences, is comprised of researchers and educators from Veterinary Sciences, Extension, the WY NASA Space Grant Consortium, the School of Computing, the WY State Veterinarian Office, the College of Education, and Molecular Biology.

The overarching mission of WyoPATH is to provide a use-inspired framework to respond to Wyoming-relevant infectious diseases. Although this framework will be broadly applicable to diseases that impact livestock, wildlife, crops, and humans, the center will focus its initial efforts on brucellosis. The overarching hypothesis of WyoPATH asserts that increased genomic surveillance and next generation adjuvants will lead to enhanced diagnostics and vaccines for brucellosis. WyoPATH success will provide enhanced food and economic security, strengthen relationships between UW and agricultural producers, and develop the next generation of the Wyoming workforce.



People in SI Programs



WRSP Scholar
Allyson Cardine
Hometown: Casper, WY
Major: Physiology

Previously, Allyson worked with Dr. Elliott Hulley, aiming to synthesize flavins to evaluate the kinetics of their oxidation chemistry with tryptamines. Allyson is currently working with Dr. Allison Mann on oral microbiome research and a survey of UW students and their Oral Health Related Quality of Life (OHRQoL). She will be presenting her research with Dr. Mann on dry mouth and a meta-analysis of the oral microbiome literature this fall at the Cold Spring Harbor Microbiome Conference in New York.



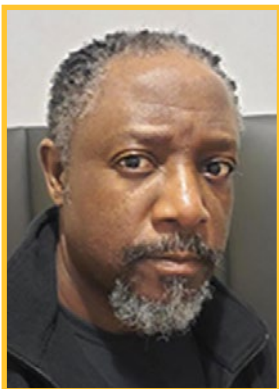
Roadshow Outreach Assistant
Erik Gwaltney
Hometown: Green River, WY
PhD candidate in Neuroscience

"I am a PhD student in the Neuroscience program using mouse models to investigate the neurocircuitry involved in circadian disturbances that occur during the progression of Alzheimer's Disease. While I was an undergraduate student here at UW, I aspired to become a medical doctor, but lab experience was required to strengthen my application. In joining Dr. William "Trey" Todd's lab, I soon discovered my passion for research and changed career trajectories. Once I started graduate school, I also decided to join the Roadshow, which simultaneously fostered my passion for teaching and pushed me to become a better educator. While I continue to study Alzheimer's and learn more about the brain, I have found that the most fulfilling part of science is sharing findings with others. The Roadshow has allowed me to do just that: I have taught physiology lessons, performed brain dissections, and presented my research with people of all ages across Wyoming, and I firmly believe these experiences have made me a better scientist and teacher. This program helps outreach assistants learn and grow as much as it helps the public, which is a truly special opportunity."



LAMP Learning Assistant
Natalie Speckner
Hometown: Lingle, WY
Major: Biology

Natalie Speckner partners with LAMP Fellow Dr. Kass Willingham to support students' active learning in General Microbiology. Natalie's teaching philosophy is one that values growth mindset. Natalie believes that true learning, "...requires a willingness to be uncomfortable." Natalie models this for her students. In fact, this summer, Natalie served as a learning assistant for the inaugural Community College Summer Institute (CCSI). Working alongside community college students from across the state, Natalie proposed solutions to housing insecurity and habitat loss in the face of environmental change. Natalie's team designed and modeled the "Hive Five Solution", a community center that contains a library, job-skills classroom, shared kitchen, aquaculture and hydroponics, and a pollinator garden. Intended to support the existing project in Casper, called Goodwill Landing, this center would ensure that previously unhoused residents would have food dignity, knowledge dignity and creative dignity. As a team leader for her students Natalie modeled what it looks like to be uncertain, to take risks and in doing so find novel, innovative solutions.



LAMP Fellow
Paddington Hodza
Instructional Professor, Wyoming Geographic Information Science Center

Dr. Paddington Hodza is a graduate of the 2025-2026 LAMP Fellows' class, Dr. Hodza envisioned, developed and implemented a novel instructional design strategy. Melding Kolb's Cycle of Experiential Learning, Problem-based Learning approaches, and Cooperative Learning structures, Paddington created a circular learning model to support authentic client-focused geovisualization work. Students worked with the Wyoming Department of Transportation's Highway Safety Program. They geovisualized school zone safety in Laramie using dashboards and storymaps. The WYDOT partner working with Paddington and his class wrote:

"We would like to extend our sincere thanks for all your hard work on the Service-Learning Project completed for the Fall 2025 Geographic Visualization class. The work you completed was groundbreaking and very impressive. The dashboard you have developed lays the foundation to significantly help the Highway Safety Program and our safety partners keep the public safe, especially in relation to school zones, which serve a vulnerable population (children)." Paddington's innovative teaching not only supports the authentic, meaningful workplace development of students, it simultaneously serves our community!